

Predicting Medical Insurance Costs Using Machine Learning

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DSC 680: Applied Data Science

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August 04, 2025

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Business Problem

The insurance companies struggle with the precise estimation of individual policyholder costs. Estimation of costs on the lower side results in financial losses, and estimation of costs on the higher side would mean unfair premium rates. This project uses machine learning models for predicting medical insurance charges given demographic and health related variables. Precise estimation would assist in accurate policy pricing, risk identification, and fairness.

Background/History

Spending on healthcare has been increasing around the world as a result of population aging, lifestyle diseases, and increasing costs of medication. The conventional actuarial approach, though common, is not likely to detect intricate patterns of data. Machine learning offers the insurer with sophisticated techniques capable of quantifying non-linear interactions among the predictor variables such as age, BMI, and smoking. The techniques will have improved forecasting potential and facilitate evidence-driven choices.

Data Explanation

The used dataset is Medical Cost Personal Dataset on Kaggle, containing 1,338 observations, seven variables, i.e., age, sex, BMI, number of children, smoking status, region, and charges. The target variable is the variable charges, and the medical costs billed for health insurance. Data preparation involved coding the categorical variables, checking for the absence of missing values, and dividing the dataset into the training set (80%) and the test set (20%).

Methods

Three models are considered. Linear Regression was employed as a baseline with the advantage of interpretability but the disadvantage of low performance for non-linear associations. Random Forest, a group of decision trees, was employed for identifying sophisticated interactions of variables. XGBoost, gradient boosting, was employed for fine-tuning performance through the use of iterative learning. Model performance was assessed in terms of R^2 score, Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

Analysis

Exploratory statistical analysis showed that charges were right-skewed, with most patients having lower charges and a small number of high-cost cases. Smoking status had the greatest impact on charges, followed by BMI, then age. Children had minimal effect, and there was minimal variation among regions.

Performance of the models showed that the Random Forest had the highest performance ($R^2 = 0.864$, RMSE = 4,591.90, MAE = 2,561.66). XGBoost also performed exceptionally well ($R^2 = 0.852$), while Linear Regression performed badly ($R^2 = 0.784$). The results above also reinforce the fact that tree models are better than linear models here.

Conclusion

Medical insurance premiums can be accurately estimated using machine learning. Random Forest was the best model, while smoking status, BMI, and age are the best predictors. These findings are in line with medical expectations and give insurers an effective tool for risk management and pricing.

Assumptions

The database also should encompass the ordinary insured population, and association variables should prove equally applicable to all population segments. Omission of fine medical history and of chronic disease should have small but not significant effects on predictions.

Limitations

The data set is small and is not populated with variables such as income, hospital visits, or chronic status that would influence predictions. The model may also reproduce the set's biases, so tight fairness assessment is required.

Challenges

The major challenges were avoiding overfitting using high-order models, giving interpretability to the stakeholders, and addressing fairness in cost estimation when the sensitive features decide the outcome.

Future Uses and Additional Applications

Future studies can also utilize true claims data for increased precision, expand the model to time-series data for monitoring cost trends, and use forecasts to guide preventive care actions.

Recommendations

Random Forest must be implemented first due to its tradeoff of performance for interpretability. More variables must be collected in an effort to be more accurate, and fairness audits must be done regularly in an effort not to become biased.

Implementation Plan

The execution entails widening the scope of data collection, training and validating the Random Forest algorithm, distributing the algorithm via an API, and ongoing performance monitoring with periodic refresh.

Ethical Assessment

Ethical issues are fairness, transparency, and privacy. The model must not be discriminatory against specific groups, stakeholders must be able to understand how the predictions are generated, and information that is sensitive must be kept secure. AI ethics guarantee that usage is responsible.

References

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